

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Grade 6 Mathematics · Standard 6.NOS.1

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINTWhat is the reciprocal of $\frac{3}{5}$?

- ☐ A $\frac{1}{3}$
- ☐ B $\frac{3}{5}$
- ☐ C $\frac{5}{3}$
- ☐ D $\frac{5}{5}$

Question 2 SELECTED RESPONSE · 1 POINTWhat is $\frac{1}{2} \div \frac{1}{4}$?

- ☐ A $\frac{1}{8}$
- ☐ B $\frac{1}{2}$
- ☐ C 2
- ☐ D 4

Question 3 SELECTED RESPONSE · 1 POINTWhat is $\frac{3}{4} \div \frac{1}{8}$?

- ☐ A $\frac{3}{32}$
- ☐ B 6
- ☐ C $\frac{8}{3}$
- ☐ D 24

Question 4 SELECTED RESPONSE · 1 POINT

What is the first step in evaluating $2\frac{1}{2} \div \frac{1}{4}$?

- ☐ A Rewrite $2\frac{1}{2}$ as the improper fraction $\frac{5}{2}$
- ☐ B Flip $2\frac{1}{2}$ to $\frac{1}{2} 2$
- ☐ C Divide 2 by $\frac{1}{4}$ and ignore the half
- ☐ D Add $2 + \frac{1}{2} + \frac{1}{4}$

Question 5 SELECTED RESPONSE · 1 POINT

What is $\frac{2}{3} \div \frac{1}{3}$?

- ☐ A $\frac{1}{2}$
- ☐ B 2
- ☐ C $\frac{2}{9}$
- ☐ D 6

Question 6 SELECTED RESPONSE · 1 POINT

A recipe needs $\frac{2}{3}$ cup of sugar. You only have a $\frac{1}{6}$ -cup scoop. How many scoops do you need?

- ☐ A 2 scoops
- ☐ B 3 scoops
- ☐ C 4 scoops
- ☐ D 6 scoops

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

What is the value of $2\frac{1}{4} \div \frac{3}{8}$?

Fill in the bubble next to the one best answer.

- ☐ (A) $\frac{1}{6}$
- ☐ (B) $2\frac{2}{3}$
- ☐ (C) 6
- ☐ (D) $\frac{27}{32}$

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Which reasoning shows why your answer to Part A is correct?

- ☐ (A) Divide only the fraction parts, $\frac{1}{4} \div \frac{3}{8} = \frac{2}{3}$, then put the whole number 2 back in front to get $2\frac{2}{3}$.
- ☐ (B) Change $2\frac{1}{4}$ to $\frac{9}{4}$. Keep $\frac{9}{4}$, change $\div$ to $\times$, flip $\frac{3}{8}$ to $\frac{8}{3}$: $\frac{9}{4} \times \frac{8}{3} = \frac{72}{12} = 6$.
- ☐ (C) Change $2\frac{1}{4}$ to $\frac{9}{4}$, then multiply straight across without flipping: $\frac{9}{4} \times \frac{3}{8} = \frac{27}{32}$.
- ☐ (D) Change $2\frac{1}{4}$ to $\frac{9}{4}$, then flip the first fraction instead of the second: $\frac{4}{9} \times \frac{3}{8} = \frac{1}{6}$.

Question 8**WRITTEN RESPONSE · 2 POINTS****Type II Reasoning: Flipping the Wrong Fraction**

Marisol divided $\frac{5}{6} \div \frac{2}{3}$. Her work was: $\frac{6}{5} \times \frac{2}{3} = \frac{12}{15} = \frac{4}{5}$. She wrote that the quotient is $\frac{4}{5}$.

Explain Marisol's misconception and give the correct quotient.

Write your answer in complete sentences. Show or explain your mathematical thinking.

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9

WRITTEN RESPONSE · 2 POINTS

Is $\frac{3}{4} \div \frac{1}{2}$ the same as $\frac{1}{2} \div \frac{3}{4}$? Solve both and explain why or why not. What does this tell you about the order in division?

Write your answer in complete sentences. Show or explain your mathematical thinking.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.